

Bidding Languages and Decentralised Trading in Markets

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Markets often feature



Heterogeneous goods



Many participants



Network structures



Bilateral negotiation



Electricity Markets



Firm-to-Firm in Emerging Economies



Digital Ad Auctions



In **market design**, we seek to

design markets with nice properties

- What agent preferences to allow?
- How can people express preferences?
- Can we compute equilibria?

understand existing markets

- Do decentralised markets stabilize?
- How desirable are equilibria?
- How robust are equilibria to shocks?

[AAAI'25]

- Propose **decentralised market dynamic** for network markets.
- Models bilateral negotiation between market participants.
- Provide theoretical and experimental insights into
 - **convergence to equilibrium,**
 - **welfare distribution,**
 - **robustness to shocks.**



with Benjamin Patrick Evans, Eleonora Kreacic, Sujay Bhatt, Alec Koppel, Sumitra Ganesh, and Paul W. Goldberg

[SSRN'25
preprint]

- Study and extend *Product-Mix Auction (PMA) bidding languages*
- Can use PMA languages in auctions and (network) markets.
 - More intuitive way to express preferences
 - Breaks down complex preferences into elementary blocks



with Elizabeth Baldwin and Paul Klemperer

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substitutes preferences

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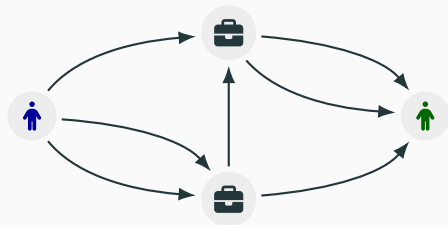


with Elizabeth Baldwin and Paul Klemperer

Decentralized Market Dynamics

Network Markets

- Heterogeneous agents I .
- Bilateral trades Ω for **indivisible goods/services**.
- Each trade $\omega \in \Omega$ represents transfer of one good/service from **buyer** $b(\omega)$ to **seller** $s(\omega)$.
- **No restriction** on underlying multi-graph of trades (cycles and parallel edges permitted).



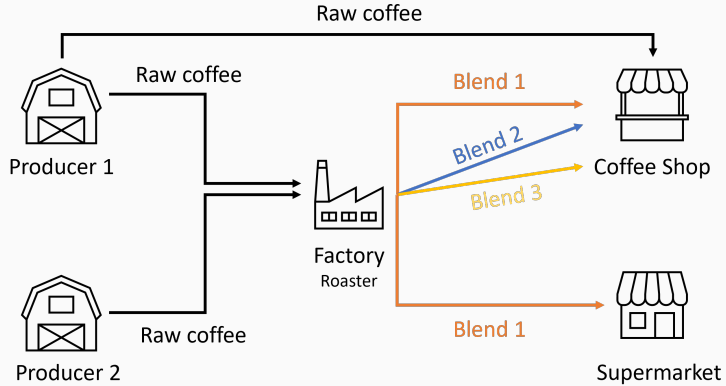
Each agent has a **valuation function** v^i mapping each bundle of its trades to $\mathbb{Z} \cup \{-\infty\}$.

Faced with prices $\mathbf{p} \in \mathbb{Z}^\Omega$, each agent demands

$$D(\mathbf{p}) = \arg \max_{\Psi \subseteq \Omega_i} \left[v^i(\Psi) - \underbrace{\sum_{\omega \in \Psi_{i \leftarrow}} p_\omega}_{\text{outcome}} + \underbrace{\sum_{\omega \in \Psi_{i \rightarrow}} p_\omega}_{\text{income}} \right].$$

Model introduced in [Hatfield et al., 2013].

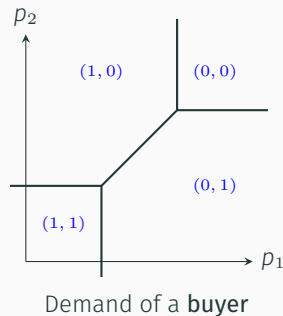
The Coffee Trade



Substitutes Preferences

Substitutes preferences are well-studied in matching [Kelso & Crawford, 1982] and auctions [Gul & Stacchetti, 1999; Ausubel, 2006].

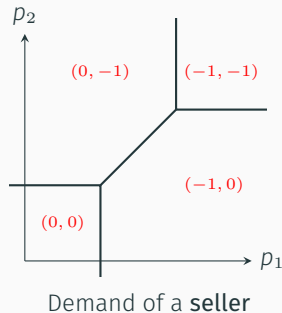
- A **buyer's** demand is **substitutes** if, for any prices p , reducing the price of some trades (weakly) decreases her demand for the remaining trades.
- A **seller's** demand is **substitutes** if, for any prices p , raising the price of some trades (weakly) decreases her demand for the remaining trades.



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Substitutes Preferences

[Hatfield et al., 2013] generalise substitutes to network markets:

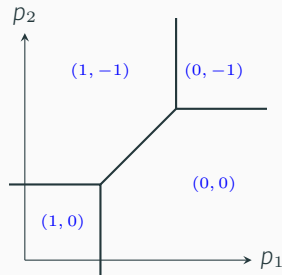
An agent's demand is **fully substitutable (FS)** if, at any p ,

(i) reducing the prices of some buying trades (weakly)

- decreases her demand for the remaining buying trades
- increases her demand of the remaining selling trades

(ii) raising the prices of some selling trades (weakly)

- decreases her demand for the remaining selling trades
- increases her demand for the remaining buying trades



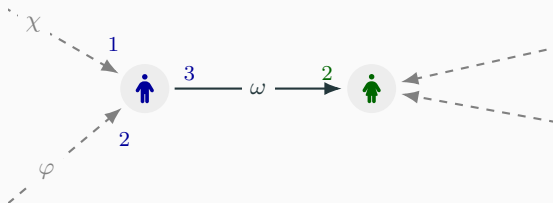
Agent buying trade 1 and selling trade 2

Theorem (Hatfield et al., 2013)

If all agents have FS demand, there exist market prices p at which all agents agree on which trades to execute.

A New Market Dynamic

- Agents i maintain an integer offer σ_ω^i for each of their trades ω .
- So each trade is associated with two offers.



The Market Dynamic

- Initialise offers σ for all trades and mark all agents as *unsatisfied*.
- While there exist unsatisfied agents:
 - Sample an unsatisfied agent i uniformly.
 - The agent observes her neighbours' offers and computes her demand Φ at these prices.
 - She updates her offers σ^i by matching her counterparts' offers for all trades in Φ and diverging by ± 1 for all other offers.
 - Mark the agent as *satisfied* and her neighbours facing modified offers as *unsatisfied*.

Equilibrium: Offer vector σ such that no agent can increase her utility by changing her offers.

Example of Dynamic



Example of Dynamic



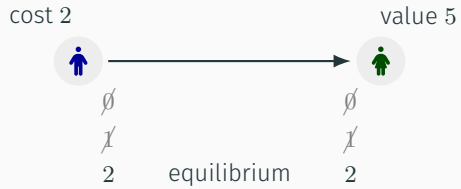
Example of Dynamic



Example of Dynamic



Example of Dynamic



Theory Results

Definition: A graph is *m-sparse* if every subgraph can be divided into two or more disjoint components by removing at most m edges.

Theorem

m -sparse markets with FS agents converge to equilibrium iff m -sparse markets with two FS agents converge.

Proposition

Markets with two FS agents and at most two trades converge.

Corollary

Markets with tree topologies converge to equilibrium (even when agents aren't FS); and 2-sparse markets with FS agents converge.

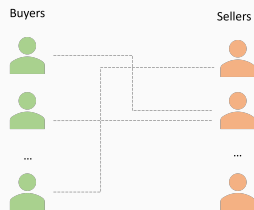
Counterexample: There exists a market with one FS agent and one non-FS agent that doesn't converge.

Experiments

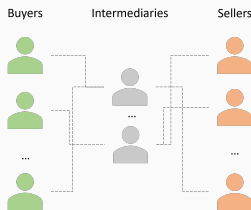
We explore our market experimentally to study **convergence rates**, the **welfare distribution**, and the **impact of exogenous shocks**.

One kind of good, and **unit-demand buyers and sellers**, and **intermediary traders**.

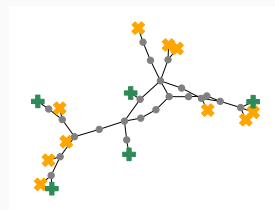
We generate three types of random networks.



BS



BIS



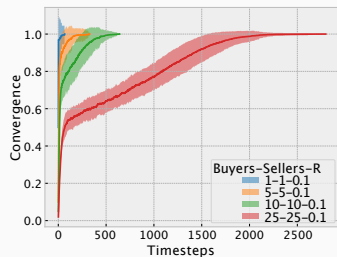
Erdős-Rényi random graphs

Result

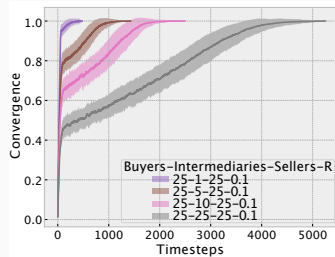
All market instances in our experiments converged to equilibrium.

Experiments | Convergence

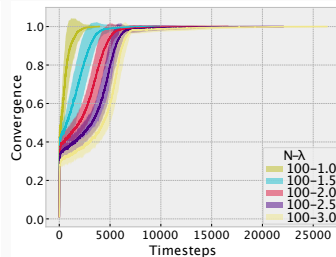
We vary the number of agents and the connectivity rate.



BS



BIS



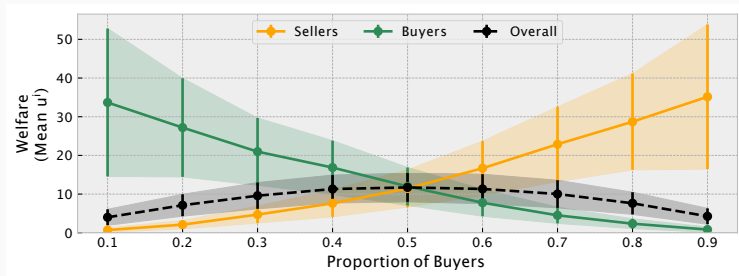
General

Observation

The number of traders and connectivity rates drives the convergence speed, due to the room for negotiation.

Experiments | Welfare Distribution

We vary the ratio of buyers to sellers.

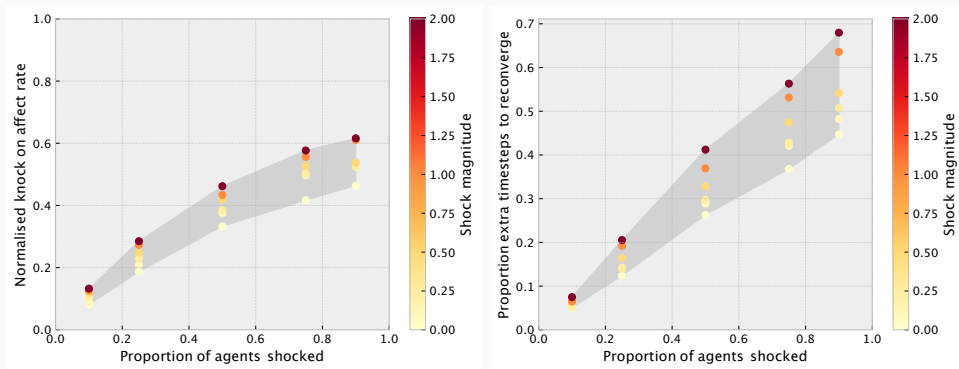


BS

Observation

Varying the ratio of buyers to sellers affects welfare distribution.

Shocks are modelled by resampling the valuations of a subset of buyers and sellers in equilibrium.



BS

Observation

The market rapidly re-converges after shocks, proportionally to the size of the shock.

We initiate the study of best-response market dynamics to understand decentralized bilateral negotiation between agents on general networks.

Our work provides theoretical and experimental evidence that our dynamic converges and performs robustly.

Future Directions

- Do markets with two agents and > 2 trades converge?
- How do our equilibria relate to centralized solution concepts?
- Develop more sophisticated market dynamics tailored to **specific markets**.
- Explore possibilities for strategic behaviour of agents.

PMA Bidding Languages

After the Northern Rock bank run in 2007-8, Paul Klemperer designed the Product-Mix Auction (PMA) for the Bank of England. Still used to allocate funds to banks.

PMA bidding languages: systematic way for participants to submit preferences before auction is run.

- The BoE's original PMA design used a simple geometric bidding language [Klemperer, 2008].
- Design has been extended in various papers [Klemperer, 2008, 2016] and studied algorithmically [Baldwin et al., 2022, 2023].
- PMA variant for the Central Bank of Iceland to buy back “offshore” accounts.

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- PMA variant for the Central Bank of Iceland to buy back “offshore” accounts.
- IMF staff interested in PMA variant to re-negotiate sovereign debt.
- Collaboration with industry partner to design PMA for trading Investment Tax Credits (ITCs).
- Consultations with Department for Energy Security and Net Zero to sell greenhouse gas removal credits with PMA.

Key considerations for bidding languages

1. Easy to understand for users, and algorithmically tractable.
2. Language should balance simplicity, expressivity and brevity.
3. Capture known classes of preferences.

New contribution

1. Representations theorems:

- *all substitutes* preferences (and no other preferences) can be expressed in the *substitutes PMA language*.
- *all strong-substitutes* preferences (and no other preferences) can be expressed in the *strong-substitutes PMA language*.

2. New, *geometric*, way of understanding substitutes and strong-substitutes.
3. Implies algorithms for learning an agent's preferences in these languages.

SSRN preprint:



Other bidding languages

- ‘Assignment messages’ [Milgrom, 2009] can’t express all strong-substitutes prefs. [Fichtl, 2021].
- Family of ‘logical languages’ by [Nisan, 2006], [Lehmann, Lehmann and Nisan, 2006] similar in spirit but not geometric. Don’t capture strong-substitutes and substitutes classes.
- Languages seem challenging for users and auctioneer to use.

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(Strong) substitutes

- Substitutes widely studied in matching, housing, labour markets.
- Strong-substitutes guarantees comp. equilibrium existence in multi-unit settings [Milgrom & Strulovici, 2009].
- Deep connections to discrete convex analysis ($M^{\natural}$ -concavity) [Shioura & Tamura, 2015].

Auction or market for trades of **bundles** $x \in \mathbb{Z}^n$ of **indivisible goods** $[n] := \{1, \dots, n\}$ available in **multiple units**.

Each agent has:

- Concave **valuation function** $v : A \rightarrow \mathbb{R}$ defined on finite set A of bundles.
- **Demand** $D_v(p) := \arg \max_{x \in A} (v(x) - p \cdot x)$.

Generalising to Multiple Units

Auction or market for trades of **bundles** $\mathbf{x} \in \mathbb{Z}^n$ of **indivisible goods** $[n] := \{1, \dots, n\}$ available in **multiple units**.

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Definition (Kelso and Crawford, 1982)

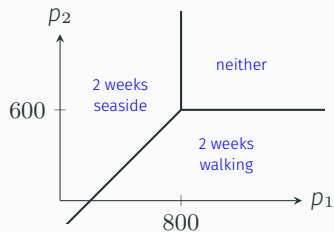
Valuation v is *substitutes* if increasing prices of some goods doesn't reduce demand in the goods with unchanged prices.

It is *strong-substitutes* if it is substitutes when treating each unit as a separate good with its own price.

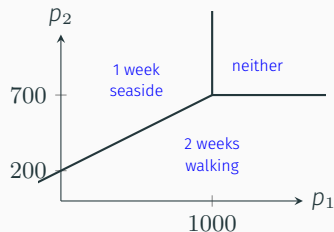
Geometric characterisation of (strong) substitutes later.

Examples

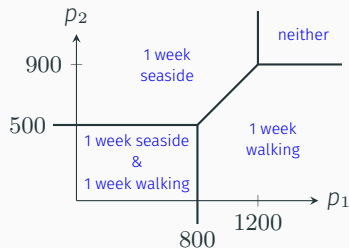
Three buyers for two kinds of holidays: **seaside** (good 1) and **walking** (good 2).



Antigone



Bathsheba



Calpurnia

In the PMA languages, every agent submits a collection $\mathcal{B}$ of *bids*.

- Each bid defines an ‘elementary’ demand.
- The participant’s demand is the ‘aggregate’ of the bids’ demands.
- More general ‘elementary’ bids lead to more expressive PMA language.

The Dot Bid language (Bank of England)

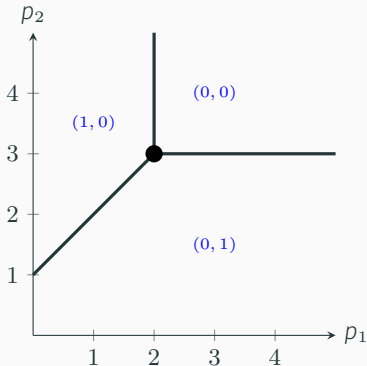
A single **dot bid** b at root $r \in \mathbb{R}_{\geq 0}^n$ expresses *unit demand*: “want at most one item.”

Vector r defines valuation

$$v(x) = \begin{cases} r_i & \text{if } x = e^i, \\ 0 & \text{if } x = \mathbf{0}, \\ -\infty & \text{else.} \end{cases}$$

At prices p :

- Bid demands *good* i maximising *utility* $r_i - p_i > 0$.
- Demands *nothing* if utilities of all goods are negative.
- In case of indifferences, break ties arbitrarily.



The Dot Bid language (Bank of England)

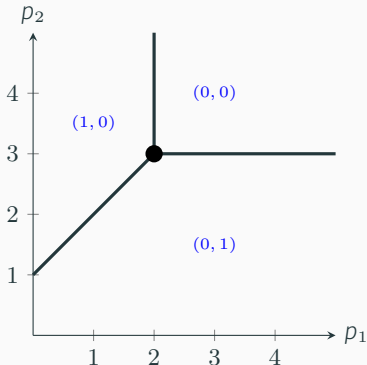
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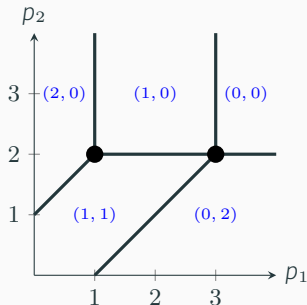
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The bid's **demand (correspondence)** is $D_b(p) := \{e^i \mid i \in J_b(p)\}$ with $J_b(p) = \arg \max_{i \in [n]} (r_i - p_i)$.

The Dot Bid language (Bank of England)

Defining the demand $D_{\mathcal{B}}(\mathbf{p})$ of a collection $\mathcal{B}$ of dot bids:



- At generic prices, $D_{\mathcal{B}}(\mathbf{p})$ is the sum of the unique bundles demanded by the bids.

$$D_{\mathcal{B}}(\mathbf{p}) := \sum_{b \in \mathcal{B}} D_b(\mathbf{p}).$$

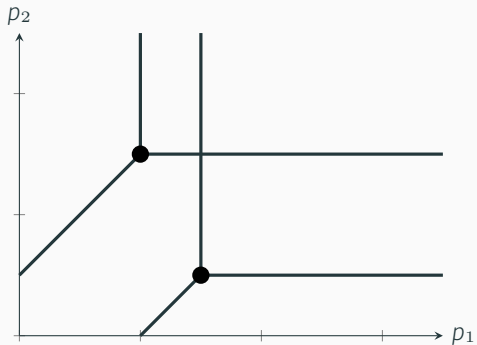
- At indifference prices, take the discrete convex hull of the aggr. demand at generic prices $Q(\mathbf{p})$ in the neighbourhood:

$$D_{\mathcal{B}}(\mathbf{p}) := \mathbb{Z}^n \cap \text{conv}\{D_{\mathcal{B}}(\mathbf{q}) \mid \mathbf{q} \in Q(\mathbf{p})\}.$$

Generalisation: bids $\mathbf{b} = (r, m)$ with *multiplicity* $m > 0$: represent m ‘unit demand’ dot bids with same root r .

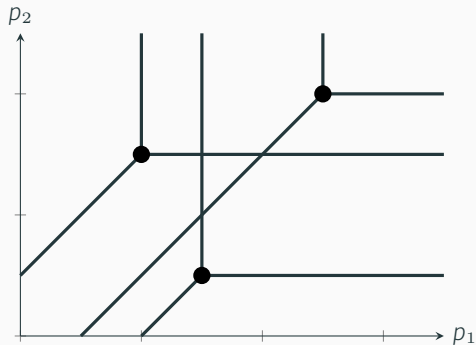
The PMA language with negative bids

So far, we can express any aggregation of unit demands.



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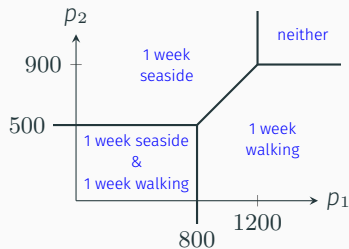
So far, we can express any aggregation of unit demands.



Easy for bidders and auctioneer.

The PMA language with negative bids

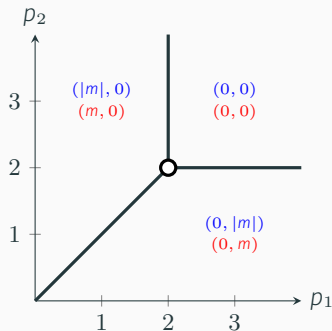
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But can't represent Calpurnia's preferences!

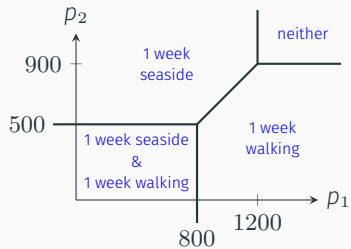
Understanding negative bids

Extension: allow negative bids (r, m) with $m < 0$.

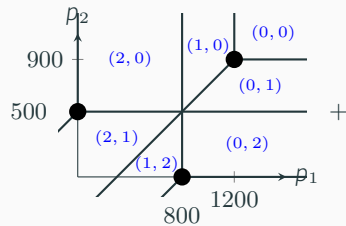
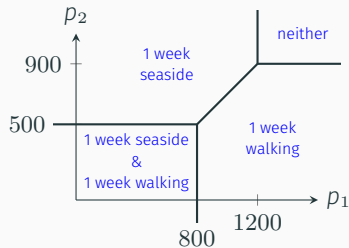


- Demand of negative bid (r, m) is negation of bid $(r, |m|)$'s demand.
- Negative bids 'remove' units from aggregation $D_B(p)$.
- NB: Demanding negative units is *not* the same as selling!

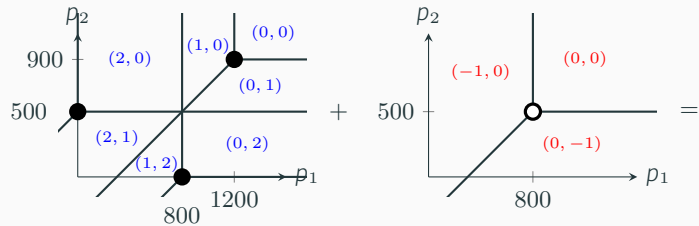
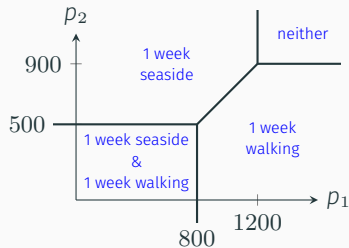
Calpurnia's demand



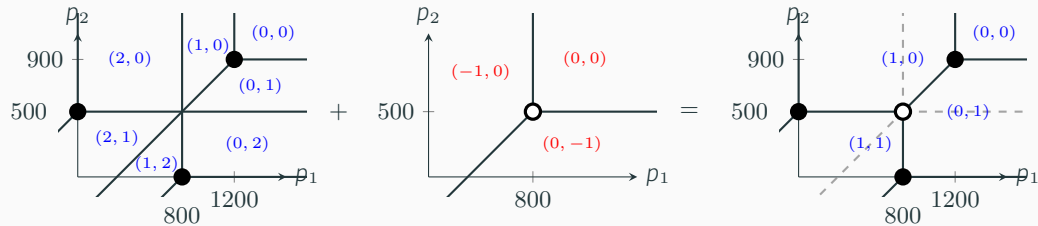
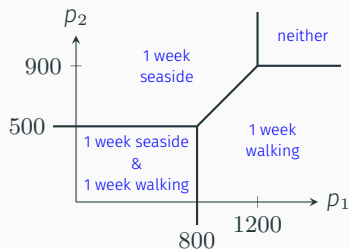
Calpurnia's demand



Calpurnia's demand



Calpurnia's demand



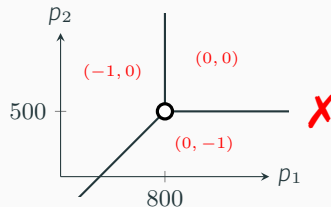
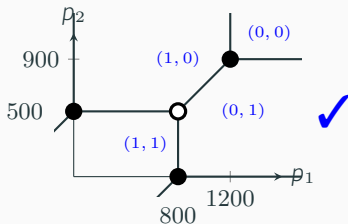
Geometrically, negative bids 'cancel out' indifference price lines.

Valid bid collections

PMA participants submit bid collections that express the demand of a valuation.

Definition

A bid collection $\mathcal{B}$ is *valid* if there exists a valuation v with $D_{\mathcal{B}} = D_v$.

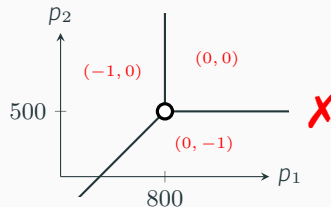
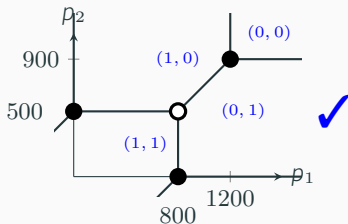


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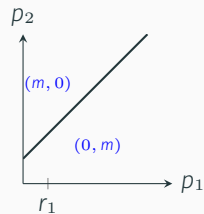
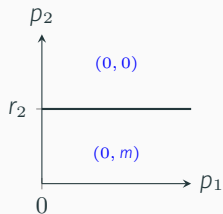
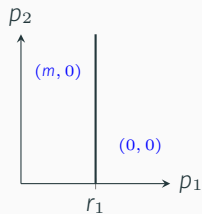
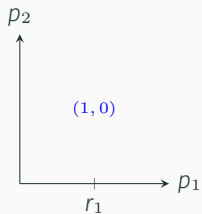


Characterisation: $\mathcal{B}$ is valid iff $D_{\mathcal{B}}$ satisfies the *law of demand*: increasing the price of one good (weakly) reduces the demand for that good.

We also develop various algorithmically checkable characterisations of validity.

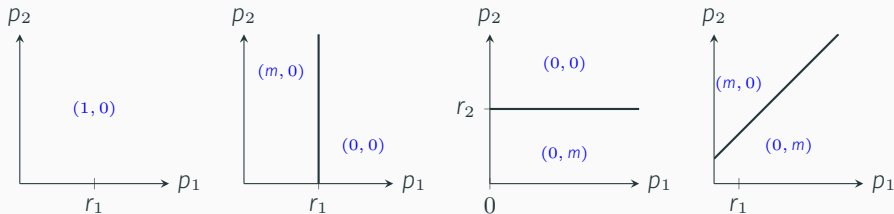
Additional building blocks

Additional 'elementary' strong-substitutes preferences:



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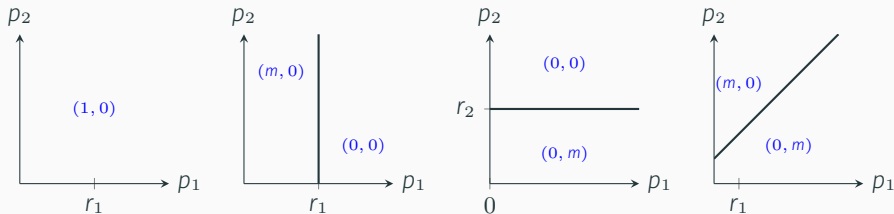


Strong-substitutes PMA language: introduce “non-regular” bids (r, m) interested only in some goods.

- Allow root entry $r_i = -\infty$ if not interested in good $i \in \{0, \dots, n\}$ at any price.
- Utility of such goods will be $-\infty$. (Definition of bid demand D_b unchanged.)

Additional building blocks

Additional ‘elementary’ strong-substitutes preferences:



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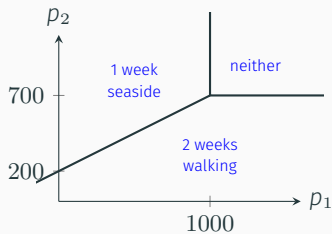
Representation theorem (second part)

Every strong-substitutes valuation v has a strong-substitutes PMA bid collection $\mathcal{B}$ with $D_{\mathcal{B}} = D_v$.

Non-regular bids important for (network) markets. Less pertinent for original PMA.

Substitutes PMA language

'Elementary' either-or demands with different quantities. E.g., "I want two apples or one banana."

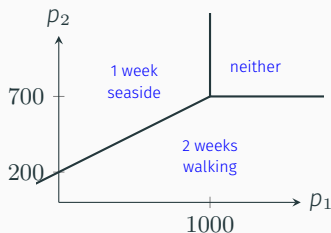


Bathsheba's bid:
 $r = (1000, 700),$

$$m = 1.$$

Substitutes PMA language

‘Elementary’ either-or demands with different quantities. E.g., “I want two apples or one banana.”



Bathsheba's bid:

$$r = (1000, 700),$$

$$t = (1, 2),$$

$$m = 1.$$

In **substitutes PMA language**, bids $b = (r; t; m)$ have additional **tradeoff vector** t capturing relative tradeoffs between good quantities.

- At generic prices, bid demands quantity mt_i of utility-maximizing good i .
- Otherwise, bid demands any integer bundle in $\text{conv}\{mt_i \mid i \in J\}$ of utility-maximising goods J .

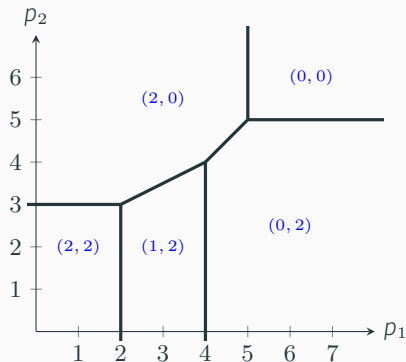
Representation Theorem (first part)

Every substitutes valuation v has a substitutes PMA bid collection $\mathcal{B}$ with $D_{\mathcal{B}} = D_v$.

The Geometry of Demand

Material from [Baldwin and Klemperer, 2019].

Locus of Indifferences Prices (LIP) $\mathcal{L}_v := \{p \in \mathbb{R}^n \mid |D_v(p)| \geq 2\}$ for valuation v .

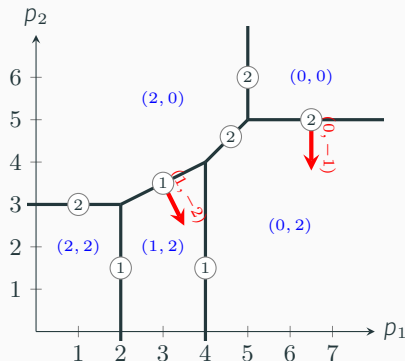


- $\mathcal{L}_v$ decomposes into $n - 1$ -dimensional linear pieces, or **facets**.
- Divides price-space into *unique demand regions* (UDRs).

The Geometry of Demand

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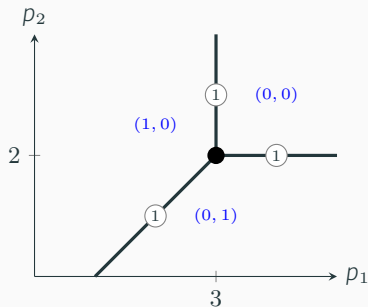
- $\mathcal{L}_v$ decomposes into $n - 1$ -dimensional linear pieces, or **facets**.
- Divides price-space into *unique demand regions* (UDRs).
- The (primitive) normal vectors $\mathbf{n}$ of each facet capture *relative* change in demand.
- Give each facet F a weight $w_v(F) > 0$ so that demand change in direction $\mathbf{n}$ is $-w_v(F)\mathbf{n}$.

Fact: The **weighted LIP** fully specifies demand changes of D_v .

We can also define weighted LIPs for bid collections $\mathcal{B}$.

First, $\mathcal{L}_{\mathcal{B}} := \{p \in \mathbb{R}^n \mid |D_{\mathcal{B}}(p)| \geq 2\}$.

Fact: For an individual (strong) substitutes bid $b = (r; t; m)$, we can write out the (strong) substitutes valuation v_b for which $D_b = D_{v_b}$.

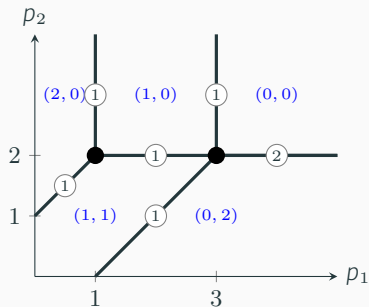


- Associate each individual bid $b \in \mathcal{B}$ with weighted LIP $(\mathcal{L}_b, w_b) = (\mathcal{L}_{v_b}, w_{v_b})$.

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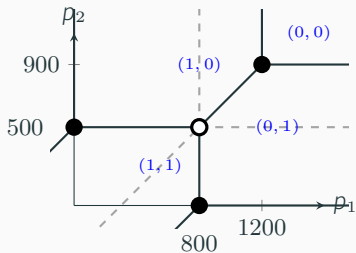


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- The weighted LIP $(\mathcal{L}_{\mathcal{B}}, w_{\mathcal{B}})$ is the ‘sum’ of the weighted bid LIPs.

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- The weighted LIP $(\mathcal{L}_{\mathcal{B}}, w_{\mathcal{B}})$ is the ‘sum’ of the weighted bid LIPs.
- When negative bids are involved, we remove any facets with weight 0 that are ‘cancelled’ out.

Fact: Demand changes of $D_{\mathcal{B}}$ are fully described by the **weighted LIP** $(\mathcal{L}_{\mathcal{B}}, w_{\mathcal{B}})$.

From [Baldwin & Klemperer, 2019] (characterisation of demand types):

- Valuation v for two goods is *substitutes* iff every diagonal facets has positive slope.
- Valuation v for two goods is *strong-substitutes* if every diagonal facets has slope 1.

The LIPs of (strong) substitutes bid collections have positive slope (slope 1) by construction, so the corresponding valuation is (strong) substitutes.

Proposition

The substitutes bidding languages only express substitutes preferences.

The strong substitutes bidding languages only express strong-substitutes preferences.

Representation Theorem

Every substitutes valuation v has a 'unique' substitutes bid collection $\mathcal{B}$ with $D_{\mathcal{B}} = D_v$.

Every strong-substitutes valuation v has a 'unique' strong-substitutes bid collection $\mathcal{B}$ with $D_{\mathcal{B}} = D_v$.

Representation Theorem | Proof Overview

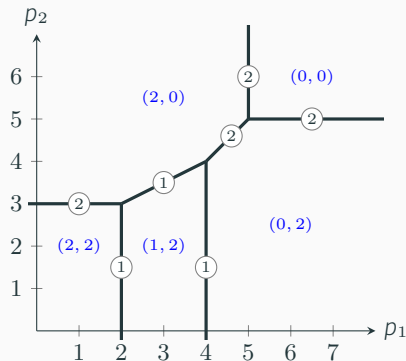
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High-level proof idea:

1. Devise procedure to build bid collection $\mathcal{B}$ from $(\mathcal{L}_v, w_v)$.
2. Prove that $D_v = D_{\mathcal{B}}$.



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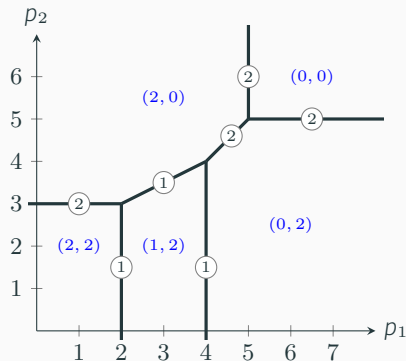
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 - 2.1 Establish $(\mathcal{L}_{\mathcal{B}}, w_{\mathcal{B}}) = (\mathcal{L}_v, w_v)$.
 - 2.2 Show that $D_v(\mathbf{p}) = D_{\mathcal{B}}(\mathbf{p})$ for some generic price $\mathbf{p}$.
 - 2.3 This implies $D_v = D_{\mathcal{B}}$.



Representation Theorem | Proof Overview

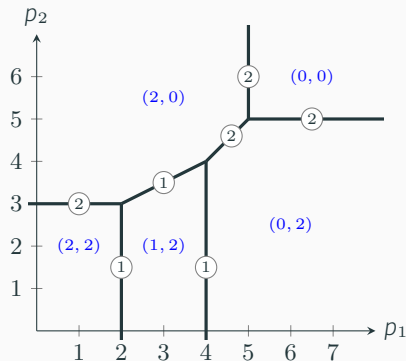
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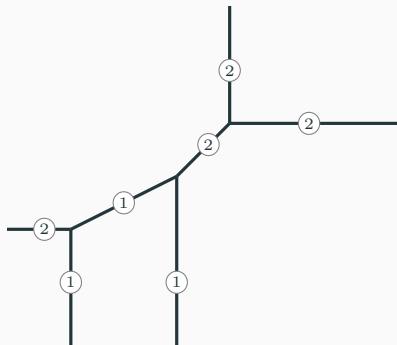
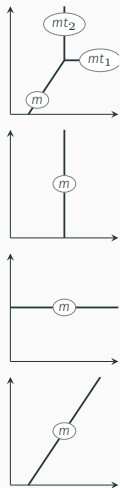
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 - 2.3 This implies $D_v = D_{\mathcal{B}}$.
3. Argue that $\mathcal{B}$ contains only strong-substitutes bids if v is strong-substitutes.
4. Show that $\mathcal{B}$ is unique.

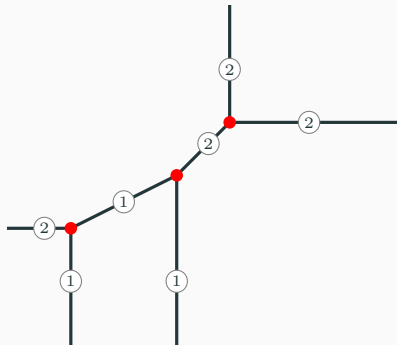
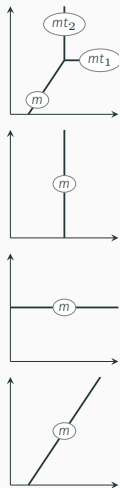


Constructing bids



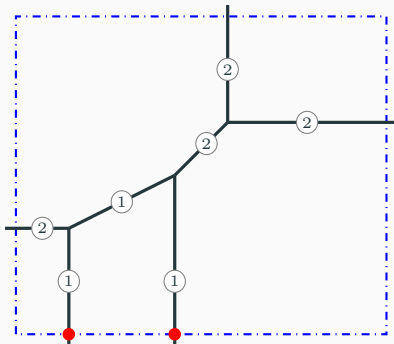
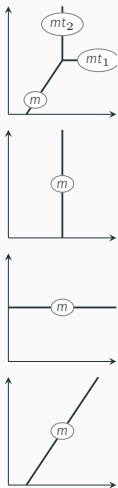
1. For every v_x of $\mathcal{L}_v$:
add **type 1** bids with matching slope and multiplicity.
2. For every intersection of lower boundary with vertical line:
add **type 2** bid with m equal to adjacent facet weight.
3. Similarly, add **type 3** bids for intersections of left boundary with horizontal line.
4. For every intersection of upper or right boundary with diagonal line:
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Constructing bids



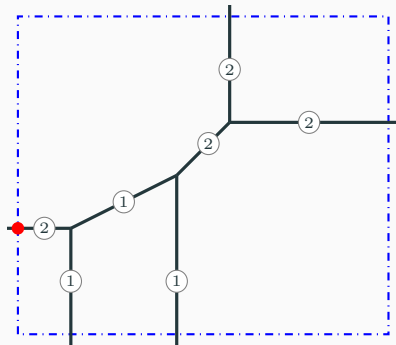
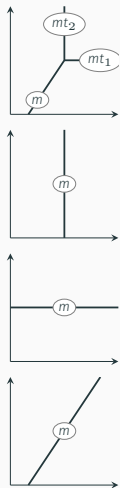
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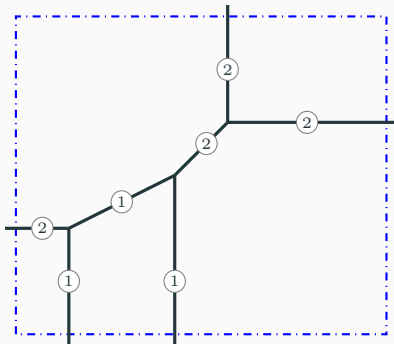
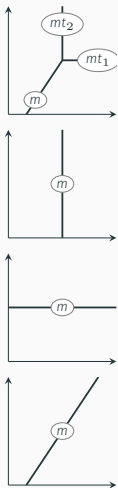
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- PMA bidding languages characterise substitutes and strong-substitutes preference class.
- New *geometric* way of thinking about these classes.
- Imply algorithms for eliciting preferences from agents using demand queries.
- Algorithmically tractable for auctioneer to compute competitive equilibrium (cf. [BGKL, 2023]), or to run decentralised market dynamic.

SSRN preprint:



Thank you!